Analog Filters Used for IIR Filter Design

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Key Concepts

1) Butterworth lowpass filters

    a) Monotone decreasing frequency response - no ripples.

    b) Maximally flat magnitude response squared near $\Omega = 0$ (first $2N - 1$ derivatives are 0 at $\Omega = 0$).

    c) Design parameters are the filter order N and the half-power frequency Ω_c.

    d) The width of the transition from passband to stopband decreases as N increases.

2) Chebyshev type I lowpass filters

a) Equi-ripple magnitude response in the passband, monotone decreasing response in the stopband.

b) Design parameters are the order N, ϵ, which determines the passband ripple, and Ω_o, which determines the passband edge frequency.

c) The width of the transition from passband to stopband decreases as N increases.

3) Chebyshev type II lowpass filters

a) Monotone decreasing response in the passband, equi-ripple magnitude response in the stopband.

b) Design parameters are the order N, ϵ, which determines the stopband ripple, and Ω_o, which determines the stopband edge frequency.

c) The width of the transition from passband to stopband decreases as N increases.

4) Elliptic lowpass filters

a) Equi-ripple magnitude response in the passband, equi-ripple magnitude response in the stopband.

b) The width of the transition from passband to stopband decreases as filter order N increases.

Video Screenshots

Discrete-time IIR filters are designed using continuous-time (analog) prototype filters

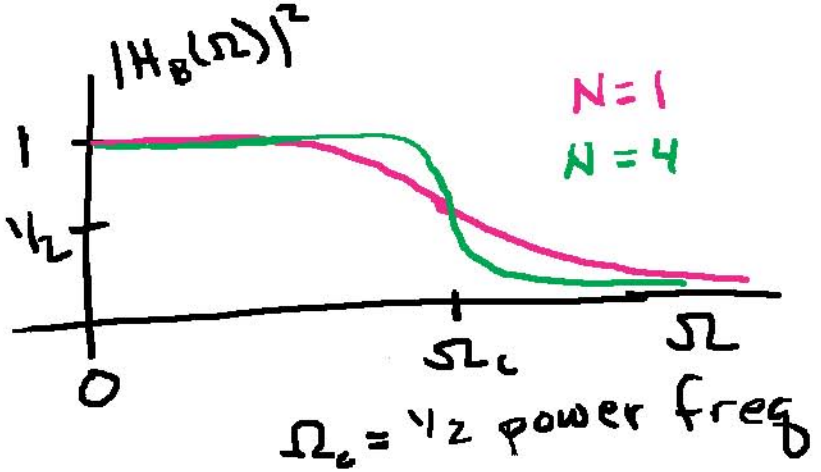
- Butterworth
- Chebyshev Type I
- Chebyshev Type II
- Elliptic

$\Rightarrow |H(s)|^2$

- design parameters
- frequency response characteristics

Butterworth filters

$$|H_B(\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$



$N \uparrow \Rightarrow$ sharper cutoff

Monotone decreasing

$|H_B(\Omega)|^2 \downarrow$ as $\Omega \uparrow$

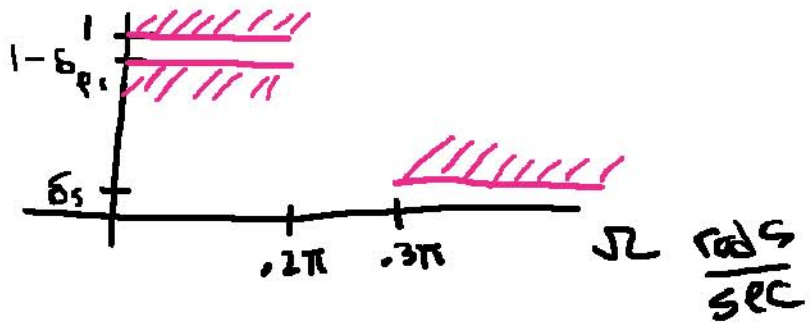
Maximally flat

For $\Omega/\Omega_c < 1$

$$|H_B(\Omega)|^2 = 1 - \left(\frac{\Omega}{\Omega_c}\right)^{2N} + \left(\frac{\Omega}{\Omega_c}\right)^{4N} - \cdots$$

1st $2N-1$ derivatives zero @ $\Omega = 0$

Design Example: choose Ω_c, N



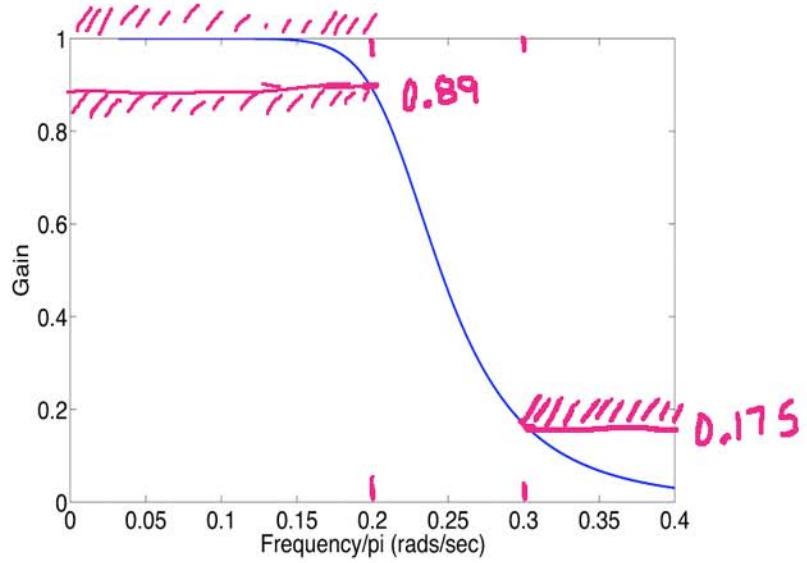
$$1 \geq |H_B(\Omega)| \geq 0.89 = 1-\delta_{p_1} \quad 0 \leq |\Omega| < 0.2\pi$$

$$|H_B(\Omega)| \leq 0.175 = \delta_s \quad |\Omega| \geq 0.3\pi$$

Response is monotone $\Rightarrow$ use band edges

pass band $$|H_B(\pi/5)|^2 = \frac{1}{1+\left(\frac{\pi}{5\Omega_c}\right)^{2N}} \geq (0.89)^2$$

stopband $$|H_B(\tfrac{3\pi}{10})|^2 = \frac{1}{1+\left(\frac{3\pi}{10\Omega_c}\right)^{2N}} \leq (0.175)^2$$



Force equality at passband edge to obtain

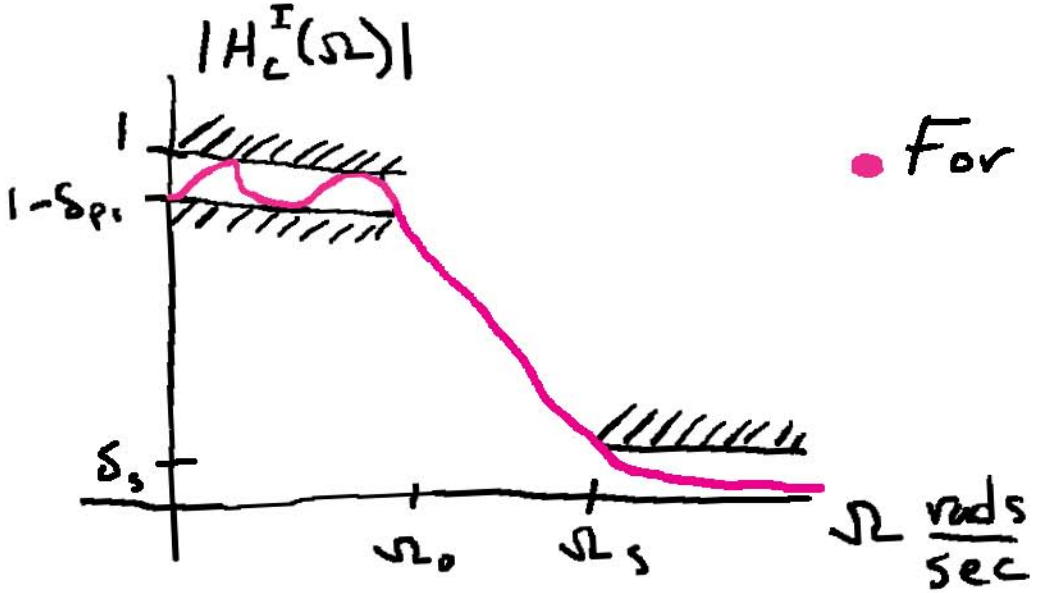
$$N = 6, \quad \Omega_c = 0.7024$$

Chebyshev Type I

$$|H_c^I(\Omega)|^2 = \frac{1}{1+\varepsilon^2 T_N^2\left(\frac{\Omega}{\Omega_0}\right)}; \qquad T_N(x) = \begin{cases} x, & N=1 \\ 2x^2-1, & N=2 \\ \vdots \end{cases}$$

- For $\left|\frac{\Omega}{\Omega_0}\right| < 1$, $T_N^2\left(\frac{\Omega}{\Omega_0}\right)$ oscillates between 0 and 1

  $\Rightarrow$ $|H_c^I(\Omega)|^2$ oscillates between $\frac{1}{1+\varepsilon^2}$ and 1



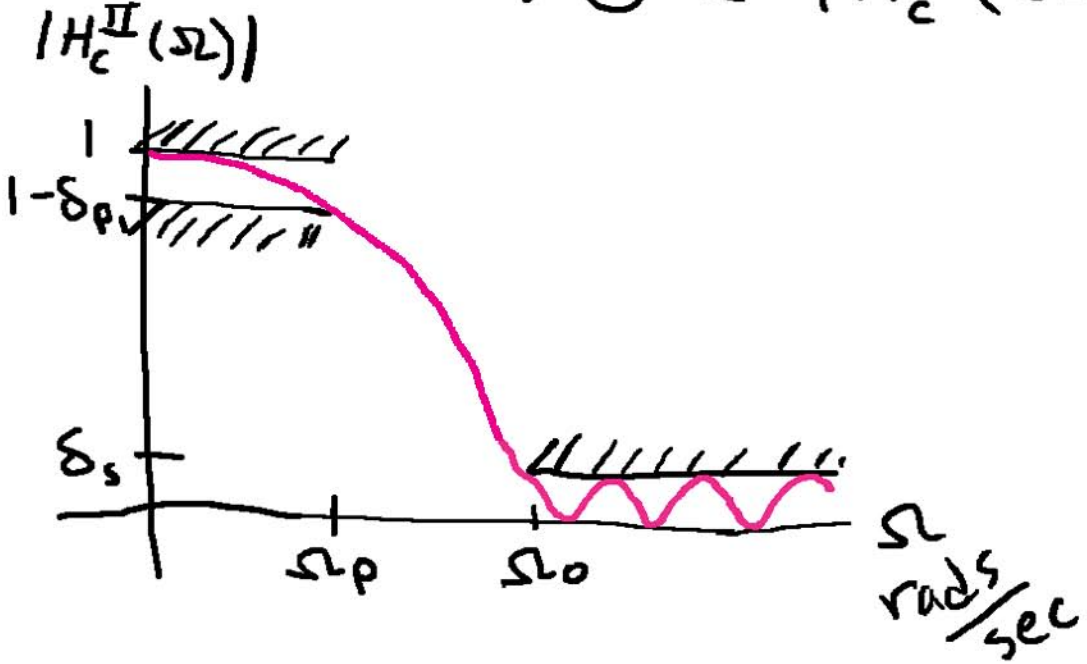
- For $\left|\frac{\Omega}{\Omega_0}\right| > 1$, $T_N\left(\frac{\Omega}{\Omega_0}\right)$ monotone increasing

Chebyshev Type II

$$|H_c^{II}(\Omega)|^2 = \frac{\varepsilon^2 T_N^2\left(\frac{\Omega_0}{\Omega}\right)}{1 + \varepsilon^2 T_N^2\left(\frac{\Omega_0}{\Omega}\right)}$$

- For $\left|\frac{\Omega_0}{\Omega}\right| < 1$ $(\Omega > \Omega_0)$, $T_N^2\left(\frac{\Omega_0}{\Omega}\right)$ oscillates between 0 and 1

$$\Rightarrow 0 \le |H_c^{II}(\Omega)|^2 \le \frac{\varepsilon^2}{\varepsilon^2+1}, \quad |\Omega| > \Omega_0$$



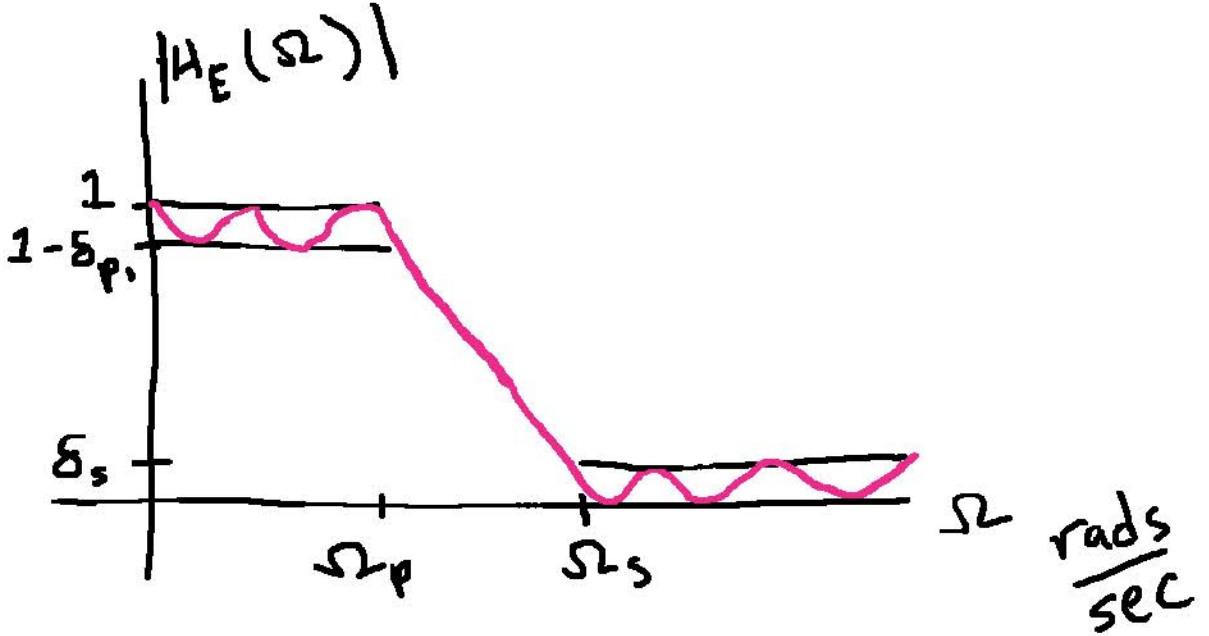
- For $|\Omega| < \Omega_0$, $T_N\left(\frac{\Omega_0}{\Omega}\right)$ increases as $\Omega \to 0$

$$|H_c^{II}(0)| = 1$$

Elliptic Filters

$$|H_E(\Omega)|^2 = \frac{1}{1 + \varepsilon^2 U_N^2(\Omega)}$$; $U_N(\Omega)$ Jacobian elliptic function

equiripple error in both pass/stop bands



Filter	Pass band	Stop band
Butterworth	monotonic	monotonic
Chebyshev I	equi-ripple	monotonic
Chebyshev II	monotonic	equi-ripple
Elliptic	equi-ripple	equi-ripple